

CIFAR10 Classification — Results Report

ResNet (from scratch) · Model Comparison & Scheduler Analysis

1. Project Overview

This report presents the results of training a custom ResNet model on the CIFAR10 dataset using PyTorch. Two configurations are compared: (A) ResNet trained with the Adam optimizer and no scheduler, and (B) ResNet trained with Adam + StepLR Scheduler. A Simple CNN baseline is also included for reference.

2. Model Comparison Table

Model	Train Acc	Test Acc	Test Loss	Epochs	Optimizer
Simple CNN (baseline)	81.06%	74.88%	0.7324	10	Adam
ResNet — no scheduler	88.34%	85.72%	0.4262	10	Adam
ResNet + StepLR Scheduler	91.18%	88.34%	0.3753	13	Adam + Scheduler

Key finding: Adding the StepLR Scheduler improved Test Accuracy by **+2.62%** over the no-scheduler version, and by **+13.46%** over the Simple CNN baseline. The scheduler also reduced the Test Loss significantly from 0.7324 to 0.3753.

3. ResNet with StepLR Scheduler — Training Curves

The StepLR Scheduler reduces the learning rate by half every 3 epochs ($\gamma=0.5$, $\text{step_size}=3$). This prevents the optimizer from overshooting the loss minimum, resulting in smoother convergence and better generalization.

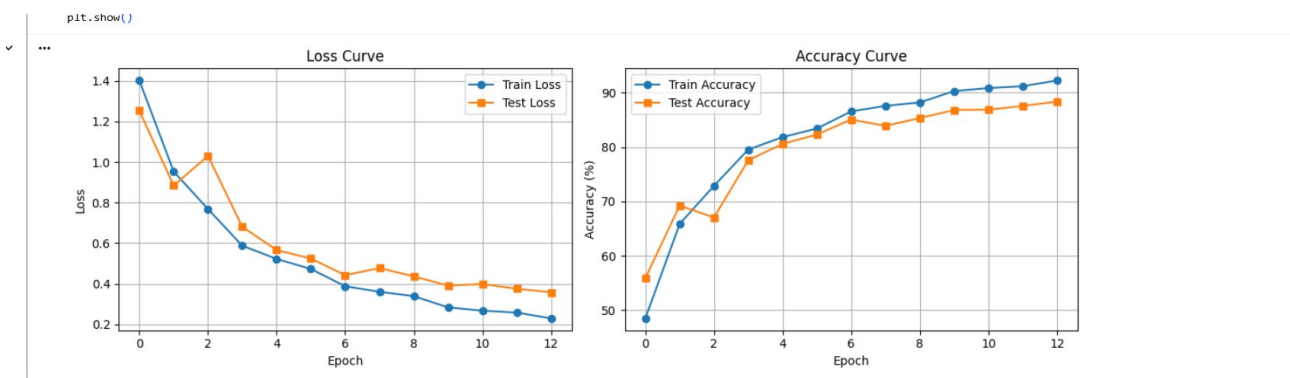


Figure 1 — ResNet + StepLR Scheduler: Loss Curve & Accuracy Curve (13 epochs)

4. ResNet without Scheduler — Training Curves

Without a scheduler, the learning rate stays fixed at 0.001 throughout training. Notice the spike in Test Loss at epoch 3 — this is caused by the optimizer overshooting the optimal region. The final Test Accuracy reached 85.72%.

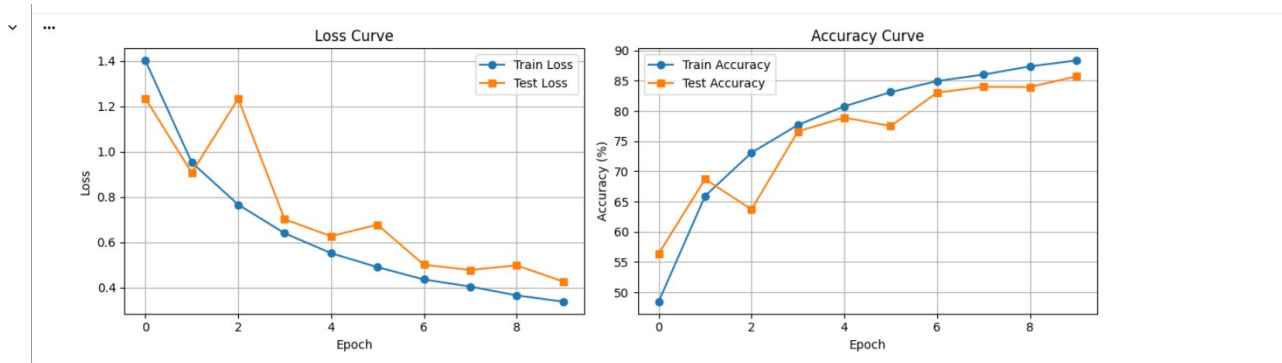


Figure 2 — ResNet (no scheduler): Loss Curve & Accuracy Curve (10 epochs)

5. Analysis & Conclusion

Aspect	Without Scheduler	With Scheduler
Test Accuracy	85.72%	88.34%
Test Loss	0.4262	0.3753
Loss stability	Spike at epoch 3	Smooth & consistent
Overfitting gap	Train-Test: ~2.6%	Train-Test: ~2.8%
Epochs needed	10	13

The StepLR Scheduler is a simple yet effective technique that removes the instability seen in the fixed-LR run. Both models show minimal overfitting (gap < 3%), confirming that Data Augmentation (RandomCrop + RandomFlip) helped generalisation. For future work, increasing epochs to 20+ with a cosine annealing scheduler could push Test Accuracy beyond 90%.